

University of Dhaka
Affiliated Engineering College
Department of Computer Science and Engineering
2nd Year 1st Semester Final Examination – 2023

Course Code: CSE 2102
Time: 3 hours

Course Title: Object Oriented Programming
Marks: 70

[Answer any Five Questions]

1. a) What is OOP? Why it is effective over structured programming? Explain. [3]
b) Write down the general form of class. Differentiate between class and objects with example. [3]
c) What is meant by data hiding? What are the differences among public, protected and private access specifier in a class? [3]
d) Write a program to find out the minimum of two given numbers. The numbers can be integer, long and double. Overload a min() function to handle those three types of data. [5]
2. a) Why do we use the keyword "static" in Java? Why constructors in Java cannot be static? [4]
b) Write a Java program to find the factorial value of a given number. [4]
c) Write short note on Garbage Collection in Java. [3]
d) What are different access modifiers in Java? Explain in short. [3]
3. a) What is method overriding in Java, and how does it differ from method overloading? What are the rules that govern method overriding? [4]
b) Why do you use super keyword? What are the conditions for a class to become abstract? Can you declare an object of abstract class? [4]
c) Create a Java class called BankAccount with properties accountNumber (String), accountHolderName (String), and balance (double). Implement a parameterized constructor that initializes these properties. Write a method to deposit money into the account and another method to withdraw money from the account and a method to display the final balance. [6]
4. a) Discuss how try-catch-finally block works in java. [4]
b) Define polymorphism. How do you declare and define an interface in Java, and what are the rules for implementing it in classes? [5]
c) What is constructor chaining? How do you call one constructor from another constructor within the same class using constructor chaining? [5]
5. a) What is inheritance? Discuss different types of inheritance with block diagram and example. [4]
b) Define constructor. How is a constructor different from a regular method? Explain with example. [3]
c) Why use java interface? Why multiple inheritance is supported in interface while it is not supported in case of class? [4]
d) Define method overloading. Can we overload main() method in java? Explain. [3]
6. a) Define generics. What are bounded types in Java generics, and how do they restrict the types that can be used as type argument? [4]
b) How can you define your own exception type in Java? Give example. [5]
c) Write a program that takes two integer numbers from keyboard to perform division operation having a try block to throw an exception when a wrong type of data is keyed or division by zero is occurred. Also write appropriate catch block to handle the exception thrown. [5]
7. a) What is the use of packages in Java? Explain the process of creating package. [5]
b) Write a Java Program to Copy the values from one object to another Object. [6]
c) What will be the output of the following Java code? [3]

```
import java.io.*;
class files
{
    public static void main(String args[])
    {
        File obj = new File("/java/system");
        System.out.println(obj.getName());
    }
}
```